SST · arXiv 2501.06749 · SAM 2 · OC-CCL

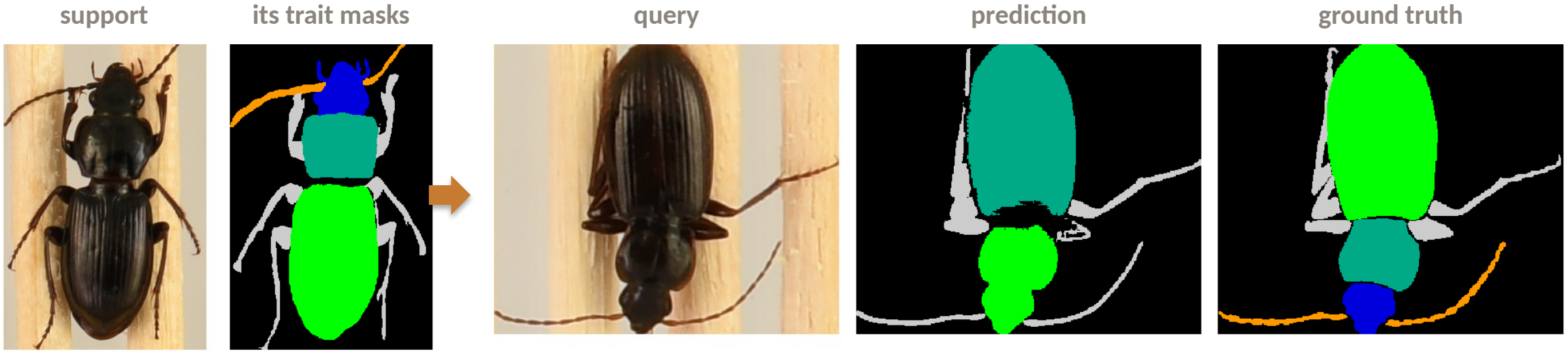
What reproduced, what didn't, and where the points actually are

Hunter Bridges · one-shot trait segmentation of museum specimens

findings to date · S14 (SAM 3 / SeC) still running

THE TASK

One labeled image. Everything else is tracking.



SST concatenates the pair into a 2-frame pseudo-video; SAM 2's memory bank propagates the support's masks onto the query.

This query is pinned the other way up — and the prediction puts the head where the support's head was.

WHY THIS REPO EXISTS

No paper-faithful evaluator existed. So that came first.

per trait

not one merged foreground blob

every subspecies

not one, chosen

20 seeded supports

model-independent, prefix-nested

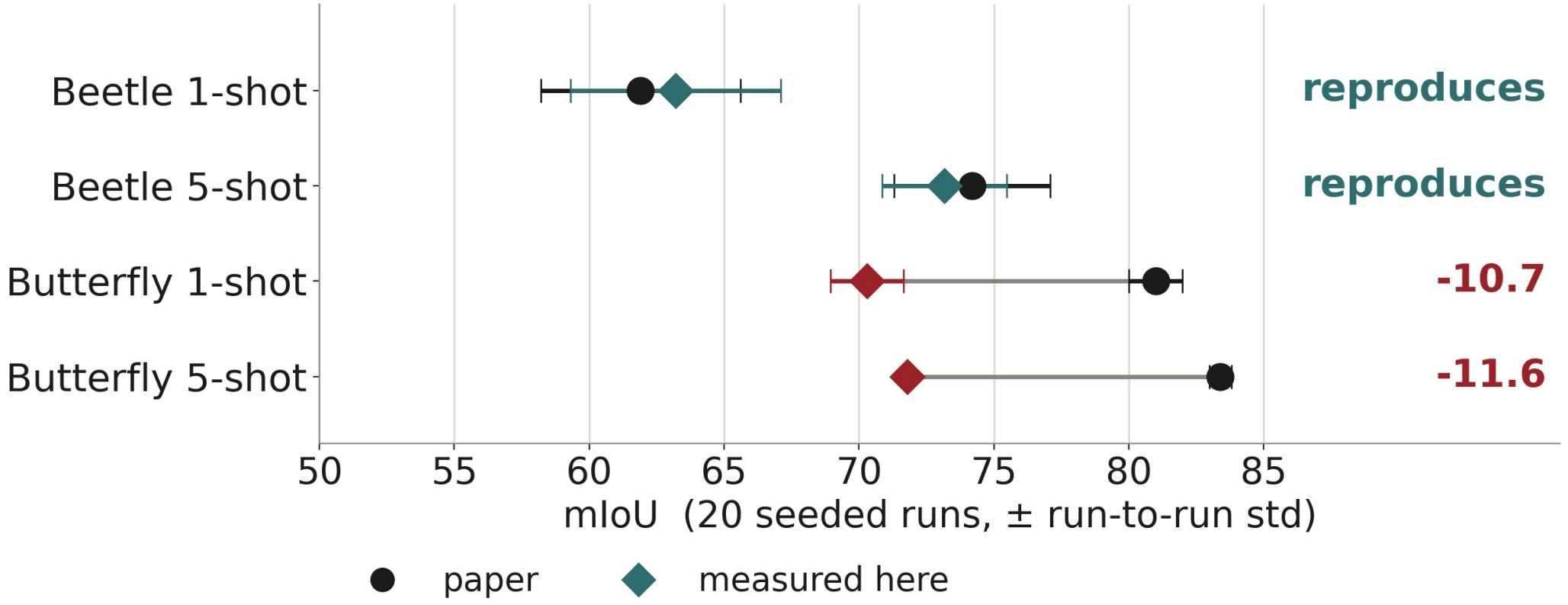
paired CIs

every arm compared seed by seed

SST's own eval script does none of these — so any OC-CCL delta measured against it is uninterpretable.

WHERE THE GATES STAND

Four reproduction gates. Two pass.

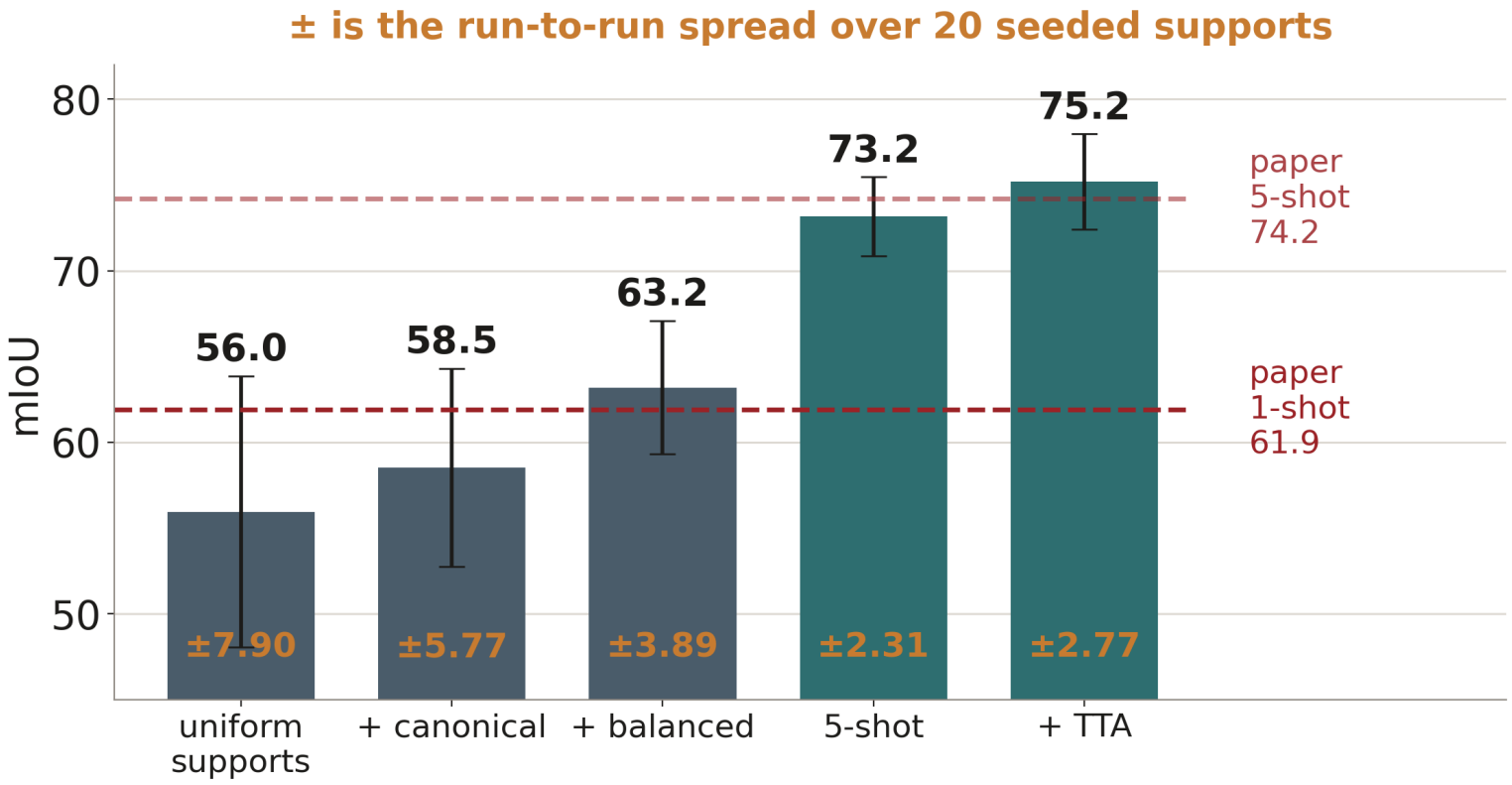


Beetle reproduces at both shot counts. Butterfly Major misses by ~10 points at both.

S4a → S4c BEETLE, 1-SHOT

A parenthetical in the paper was the whole 1-shot story.

“one labeled image (with canonical shapes and visually clear traits) is sampled as the reference”



THE RULE THIS PROJECT RUNS ON

Check the spread before the mean.

7.90	5.77	3.89	3.70
uniform supports	+ canonical	+ species-balanced	the paper

The std measures sensitivity to support choice, and is invariant to which supports were drawn.

The mean drifts by the SEM even with a correct implementation — the paper's 20 reference IDs were never published.

S5 BUTTERFLY MAJOR, 1-SHOT

Butterfly Major: 70.30 against 81.0 — and eight causes ruled out.

70.30

measured

81.0

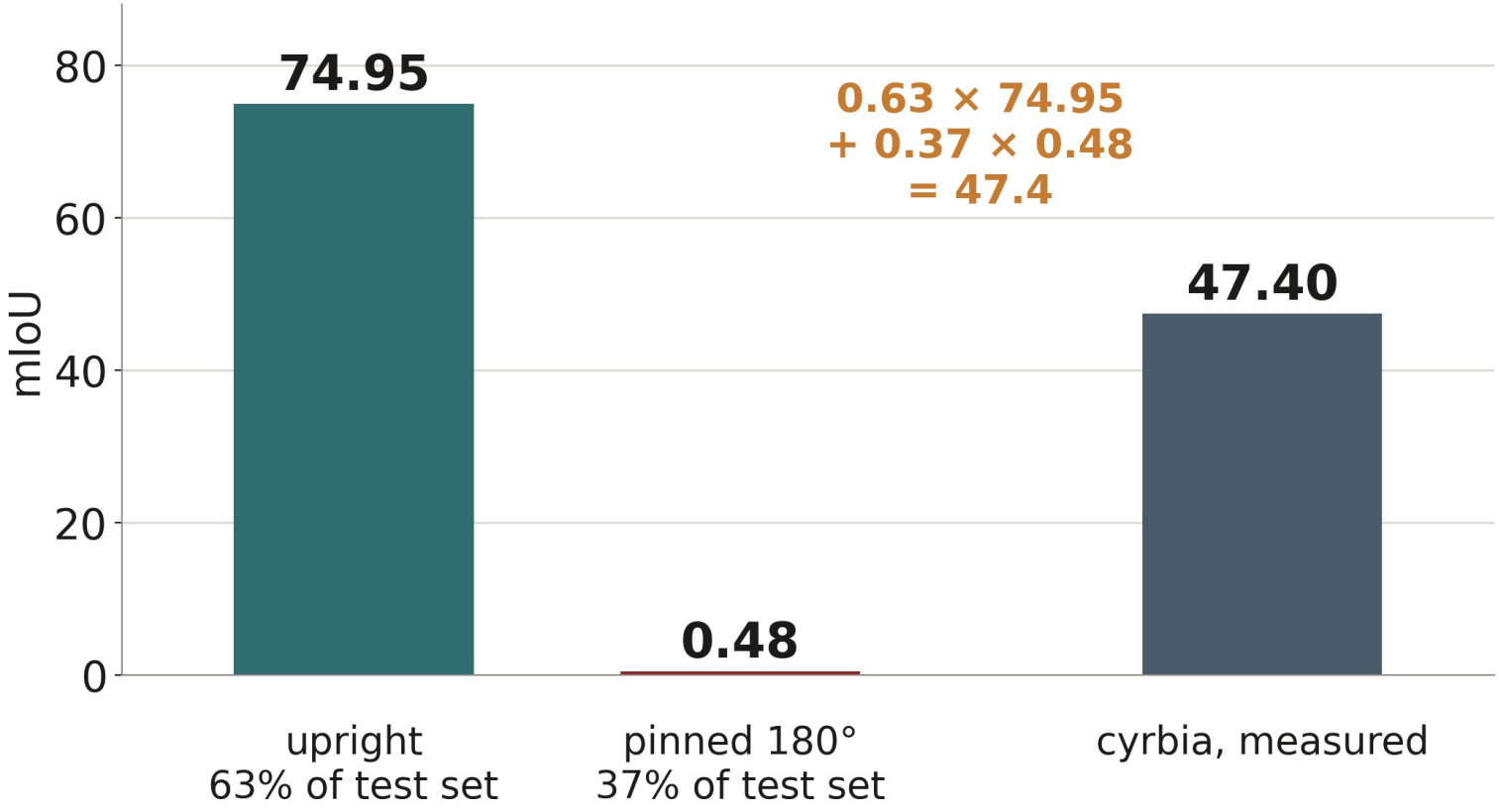
the paper

- metric aggregation
- IoU space
- support-absent traits
- masks · splits · resize
- support completeness
- stratum balance
- the overlap rule
- image resampling

The copy baseline — support label map pasted on, no model — reproduces the paper's Table A3 to 0.14, so masks, splits, vocabularies and resize are all exonerated.

THE SPECIFIC DEFECT

37% of one subspecies is pinned upside down.



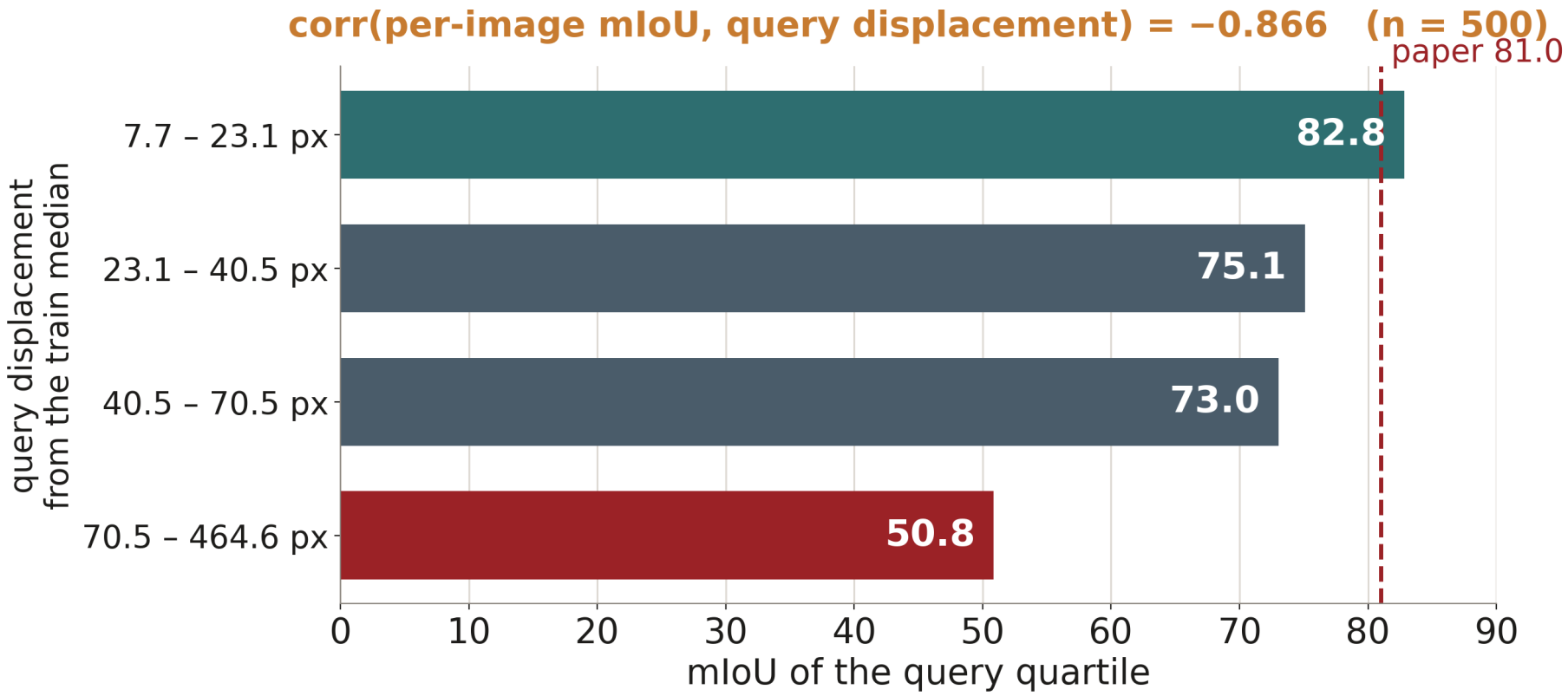
The copy baseline scores 0.00 on exactly those specimens.

No model is involved — so this is a property of the data, not a SAM 2 behaviour.

The paper reports 81.0 on a test set that contains this subpopulation, and describes no canonicalisation step.

THE GENERAL DEFECT

The model transfers spatial layout, so the query's arrangement decides the score.

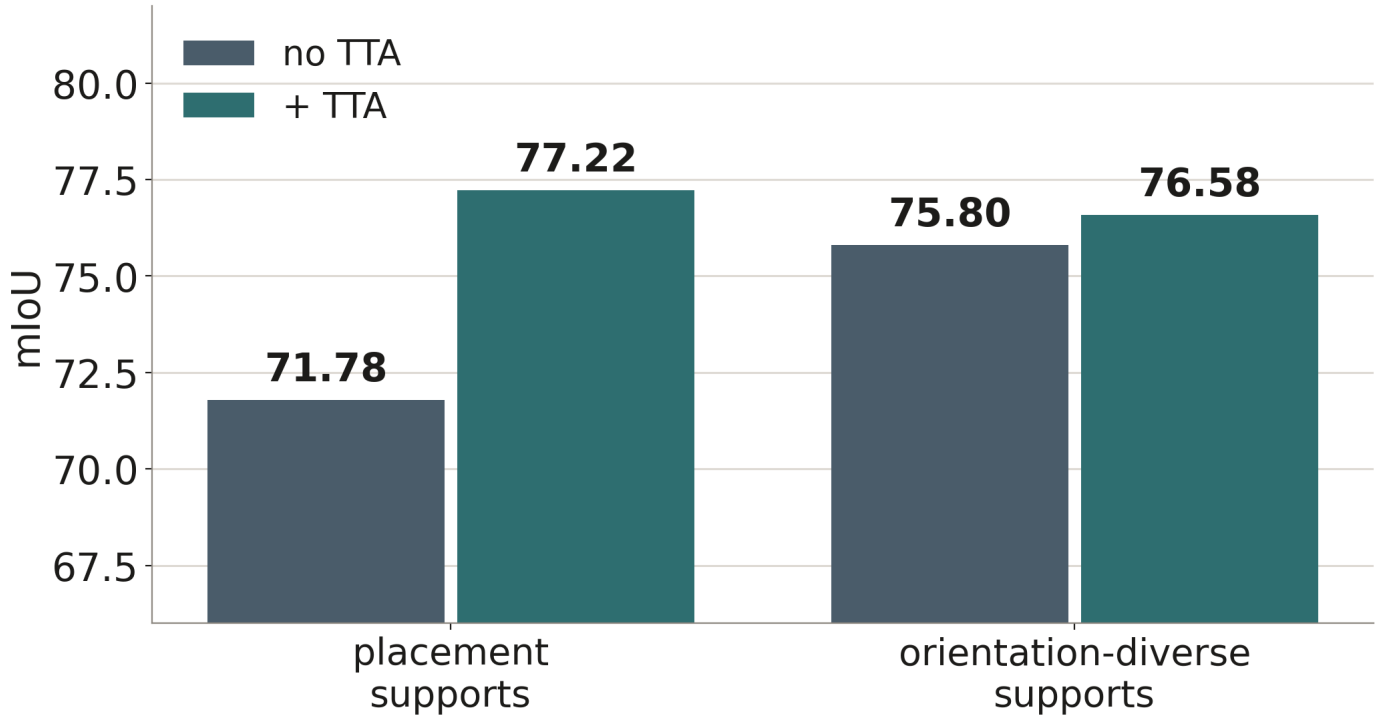


On well-arranged queries we already exceed the paper. The whole deficit lives in the badly-placed tail — and the test set cannot be curated.

S6c THE ROTATION 2×2

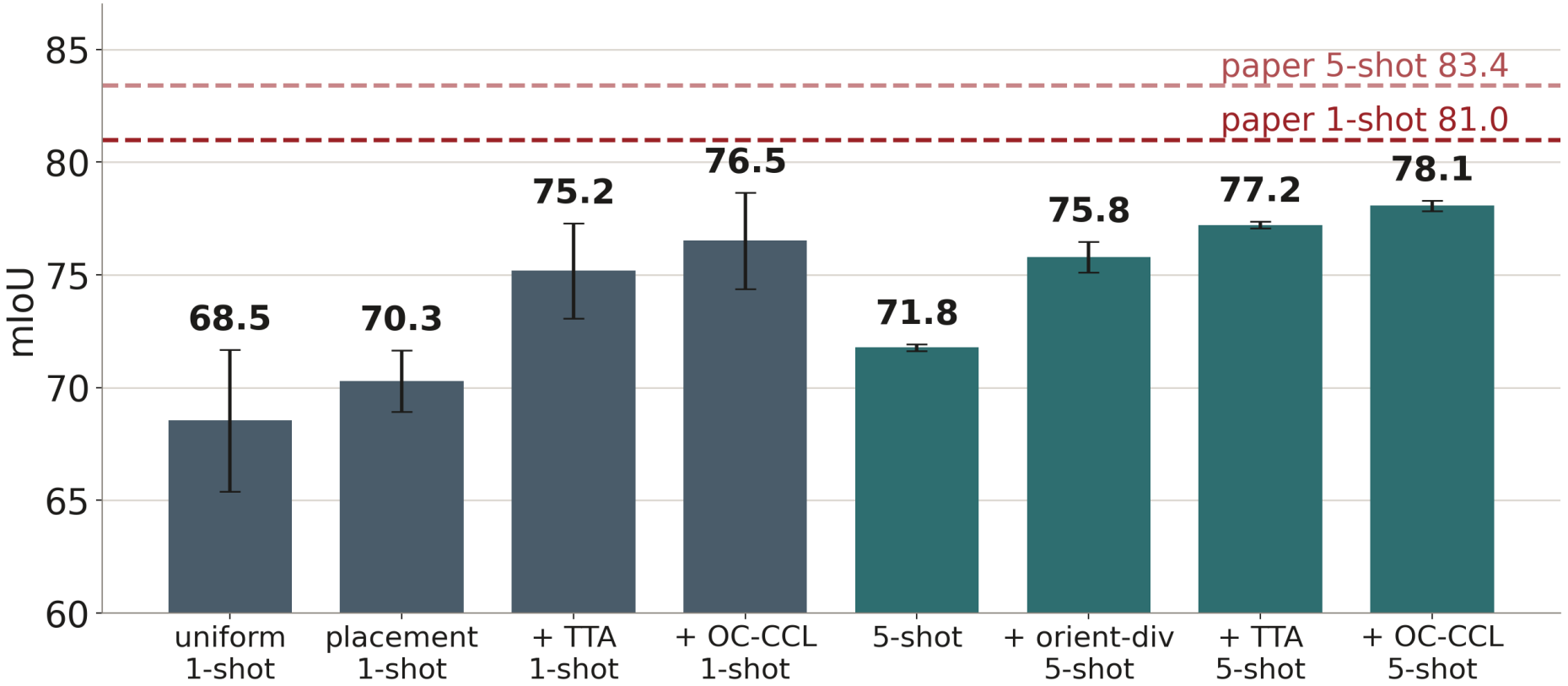
Two fixes. One defect. Combining them costs score.

support-side +4.02 · query-side +5.44 · both +4.80 · interaction −4.66



Support-side (draw both orientations) and query-side (predict twice, keep the confident one) buy the same +4 to +5.

What each technique bought.



Best measured: 78.07. Still 5.3 short — and ~5 of that sits below the silhouette ceiling, where no assignment-side method reaches.

PART TWO

The loss

OC-CCL: one-shot cycle-consistency loss.

The paper reports +3.3 mIoU on Beetle.

S8a THE FIRST DEFECT

The released cycle prompt severs the gradient.

```
soft_query_mask = (pred_query_mask > 0).float()
```

0.00

forward-leg gradient, as released

26× / 351×

forward ÷ reverse after the fix (merged / per-trait)

The fix is exact, not an approximation: SAM 2 maps a prompt mask to logits with fixed constants, so feeding (logits + 10) / 20 reconstructs them identically and the path becomes linear.

The paper says the prompt “remains differentiable”. The released code is what deviates — the first of five such axes.

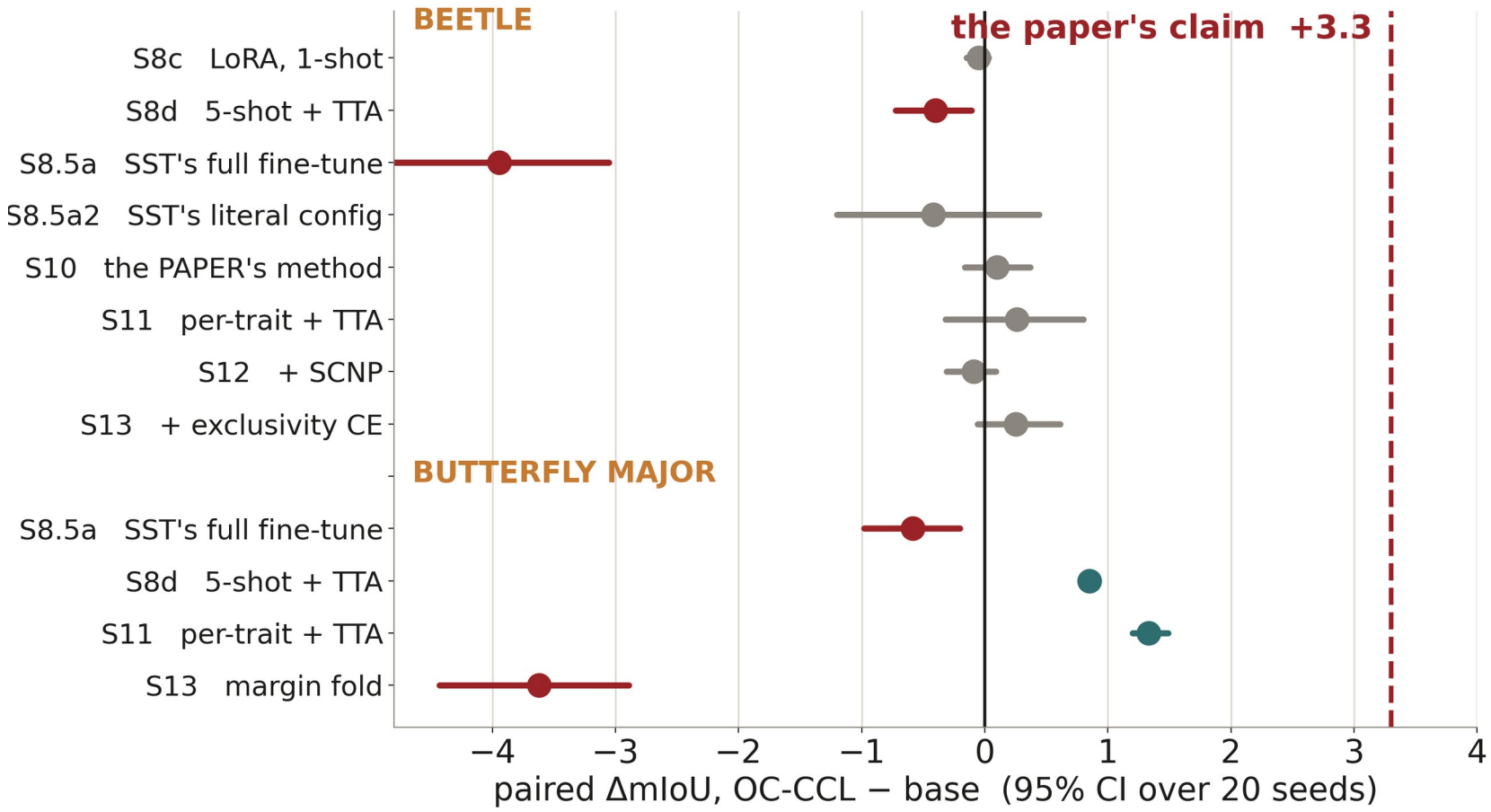
The released code contradicts its own paper on five axes.

	THE PAPER	THE RELEASE
cycle prompt	“remains differentiable”	(pred > 0).float()
supervision	C distinct trait channels	one merged foreground
method	LoRA r=32, α=64	full fine-tune, LoRA off
epochs	“train it for one epoch”	10 epochs
labels	one fixed labeled image	404 – 4,227 labels

The OC-CCL code landed ~16 months after the paper, from a different contributor. So “does OC-CCL reproduce?” had to be asked twice.

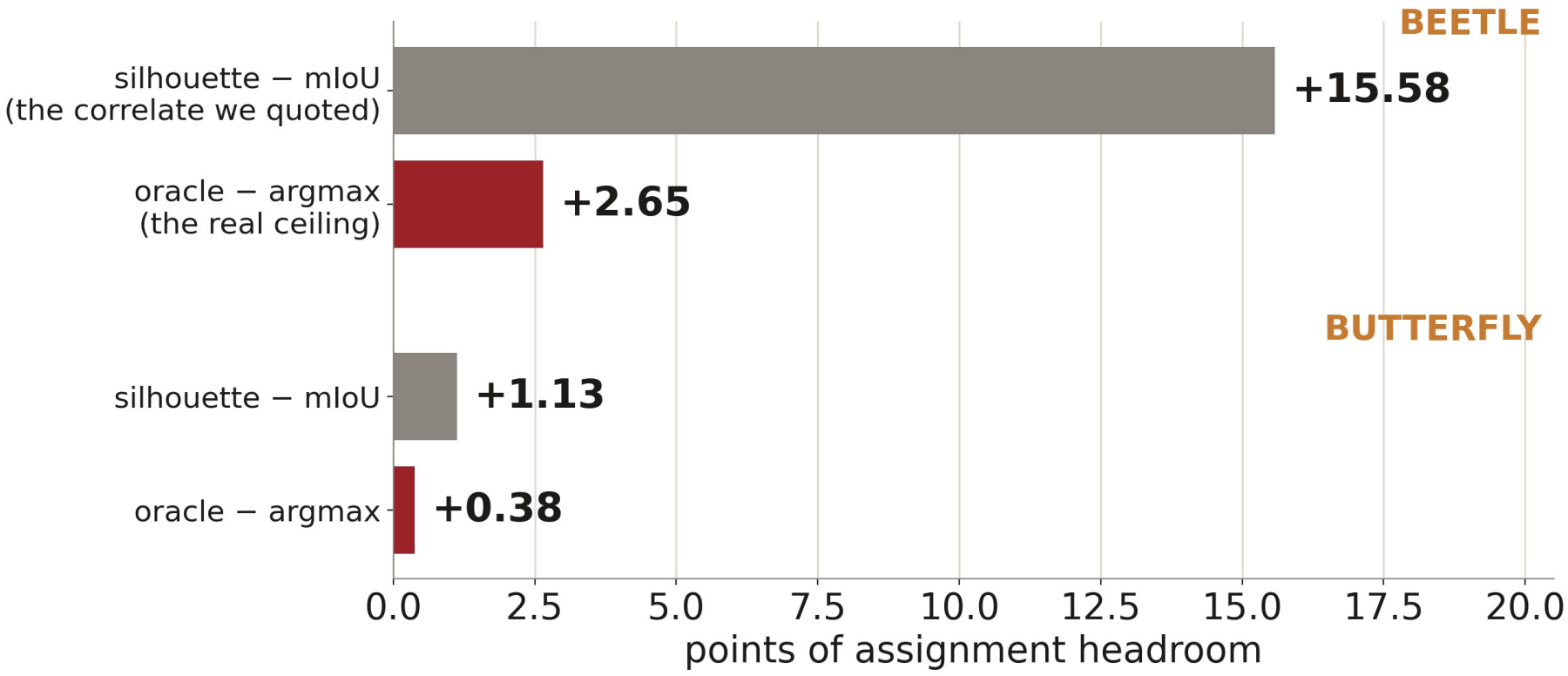
S8 → S13 EVERY OC-CCL CONFIGURATION WE COULD RUN

The claim is +3.3. Nothing we measure is near it.



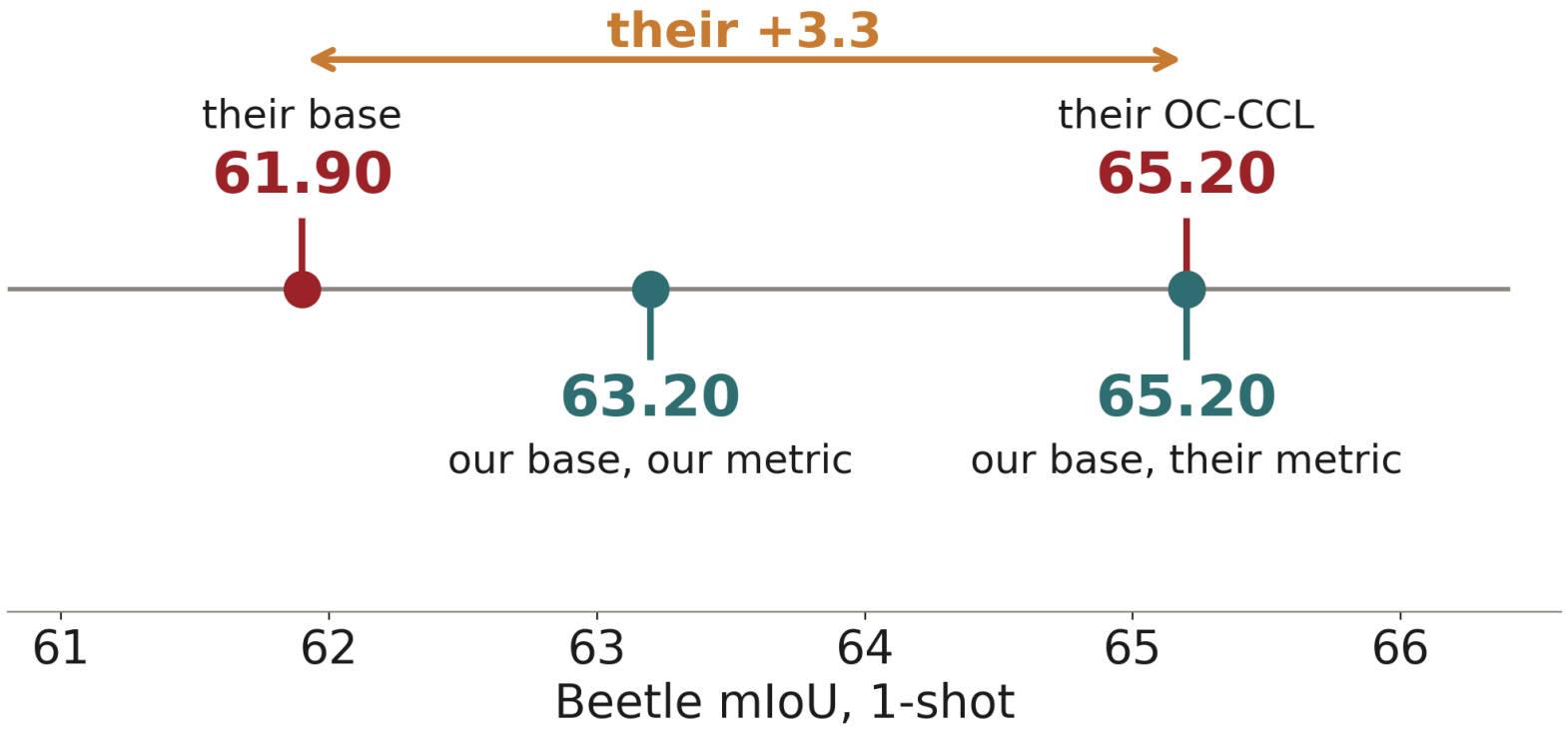
Release config, paper config, three replicate references, per-run anchoring, both label budgets, ±augmentation, both overlap rules, five theory-selected loss variants.

We were measuring a correlate, not a ceiling.



The loss arms had been competing for 2.65 points, not 15.6. The rest is detection failure — regions the tracker never proposes, which no loss on proposed logits can reach.

Under the paper's own metric, every number locks together.



Their pipeline ran ~3 points below ours; their OC-CCL recovered roughly that deficit on their pipeline; on ours there was nothing left to recover. Caveat: 65.20 vs 65.2 is a coincidence-grade match at SEM ≈ 0.8.

WHAT THE PROJECT SUGGESTS

Four things, in order of how much they cost to learn.

1 Unpublished protocol decides the number

Six evaluation details were never published — reference IDs, the beetle split, the pair code, the LR, the overlap rule, the boundary tolerance. Curation alone is worth +7.2.

2 A delta needs a pairing and a band

Every arm here is paired per seed with a CI and a band frozen before the run. That is what turns “no effect” into a result instead of a failed experiment.

3 Measure the ceiling; never infer it

“+15.6 headroom” was a correlate. The measured ceiling was 2.65, and it retired an entire line of work.

4 The remaining points are model capacity

Support-side and inference-side levers are exhausted. ~5 of butterfly's 6 points sit below the silhouette; beetle's residual is unproposed regions.

S14 RUNNING NOW

The swap the loss programme was built for.

SAM 3

A newer tracker that keeps SST's mechanism wholesale.

Geometry pinned at 1008 / 1152 / 7, strict 0-0 weight load.

SeC-4B

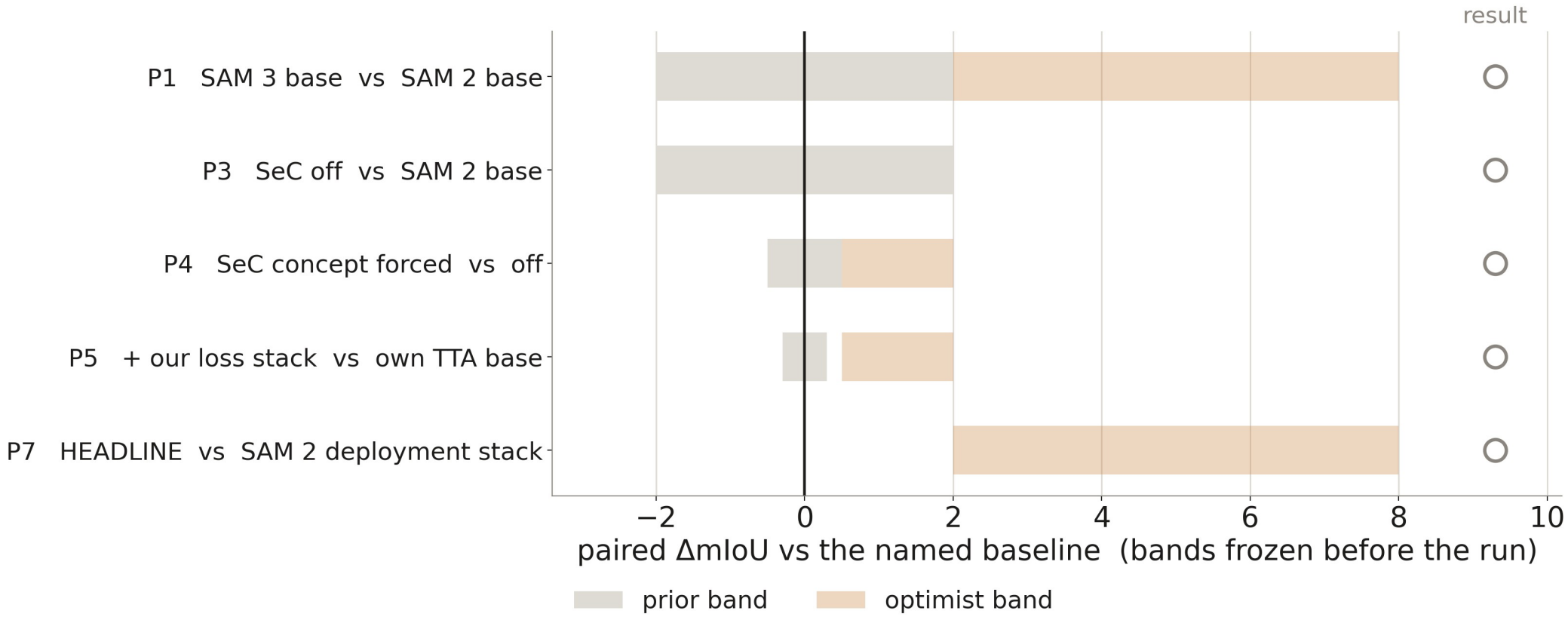
Concept-level identity on top of a SAM 2.1 tracker — the one mechanism that could reach the 180° cohort that no pixel memory does.

Held fixed: the same 20 seeded draws, each dataset's own best 1-shot protocol, and the crowned stack — per-trait OC-CCL + exclusivity CE + TTA.

Cross-family rows are the comparison the comparability key exists to refuse. They go through one sanctioned relaxation, and every row carries the fields it ignored and both their values.

S14 PRE-REGISTERED, PENDING

The bands were frozen before the GPU ran.



Results drop into the open circles. The gates print the deltas against these bands and never gate on sign.

P2 decides which story S14 tells.

oracle – argmax > ~+5

The new backbone proposes what SAM 2 never did, and assignment now lags behind the proposals.

The assignment pot is real again — and the loss programme, closed on SAM 2, reopens on the new family.

oracle – argmax ≈ +2.65

Same proposal boundary as SAM 2 — the upgrade is cosmetic on this benchmark.

The residual is a property of the data and the task, not of the tracker generation.

P7, the headline: stack + TTA vs SAM 2's 76.52 / 64.12 → _____

Either reading is a finding. Both are written down before the numbers exist, which is the only way that stays true.

WHAT STAYS UNFALSIFIABLE

Six things about the benchmark we cannot check.

- the 20 reference image IDs
- the Beetle train/test split
- the Beetle pair-construction and eval code
- the fine-tuning learning rate
- how overlapping trait predictions are resolved
- the one-pixel boundary tolerance

Our beetle split is synthesized; our curation approximates a human judgement the paper never defines. Both are reported as such, and all three support arms are always shown.

WHERE IT STANDS

Beetle reproduces at both shot counts.

Butterfly Major does not — about half of that gap is orientation, and most of the rest is mask quality.

OC-CCL's +3.3 is not recoverable from the code, the paper, or the metric ambiguity.

The next points live in the model. That run is on the GPU now.